

Week 8 Knowledge Check — AI Spec Development & Capstone

A short retention check covering the week's core ideas: the AI Spec Development pattern, capstone discovery and the compressed PRD, compressed architecture (TCD/TMD-light), technical-logic validation and finalization, and Friday's presentations. Answer each question, then check yourself against the key at the end. Aim for concepts, not trivia — every answer maps back to something we used in the capstone.

Format: 12 questions (multiple choice + true/false). ~15 minutes. Individual or triad.

Questions

1. **(MC)** In the AI Spec Development pattern, the 5-prompt sequence is run how?
 - A) All 5 prompts in parallel, then validate the assembled spec once at the end
 - B) Each prompt validated before the next runs — a validation chain
 - C) As one mega-prompt that includes every input at once
 - D) Prompts 1–4 draft the spec; prompt 5 validates all four together
2. **(T/F)** Senior TPMs use AI Spec Development to draft the spec, not to do the thinking — judgment shapes every section and the AI accelerates the typing.
3. **(MC)** In the capstone, "compressed" work means:
 - A) Lower-quality work produced faster
 - B) Selective work — keep what's load-bearing, cut what's cosmetic
 - C) Only the happy-path sections, skipping edge cases
 - D) AI-generated drafts run with lighter validation
4. **(MC)** In the compressed PRD-light, the acceptance-criteria coverage target is:
 - A) 6 happy / 0 sad / 0 weird
 - B) 3 happy / 2 sad / 1 weird
 - C) 2 happy / 2 sad / 2 weird
 - D) 5 happy / 1 sad / 0 weird
5. **(MC)** When running the AI Spec sequence under time pressure, which content must be preserved verbatim rather than paraphrased by the AI?

- A) The engineering-ready summary paragraphs
- B) Trade-off costs, revisit triggers, named invariants, and customer quotes
- C) The headline and the section titles
- D) The provenance log entries

6. (T/F) In the Day-3 cross-triad spot-check, a reviewer triad that reads the AI Spec and reports "looks fine" without checking claims against the source has done its job correctly.

7. (MC) The 7-point technical-logic validation on Day 4 exists mainly to check:

- A) That each artifact is well-written on its own
- B) Whether the artifacts fit together and cohere as one set
- C) That the AI Spec matches the presentation deck word-for-word
- D) That grammar and formatting are consistent across documents

8. (T/F) Most triads pass all 7 technical-logic checks on the first pass.

9. (MC) In the cross-artifact integration pass, a good cross-reference must be:

- A) Short and informal — "see the TCD" is enough
- B) Specific (cite the Section), verified, and add value
- C) Placed only in the AI Spec, not the other documents
- D) Generated by AI to guarantee consistency

10. (MC) On the Friday presentation rubric, which dimension carries the single highest weight?

- A) Customer + problem clarity (15%)
- B) AI Spec quality (20%)
- C) Trade-off honesty (15%)
- D) Presentation craft (15%)

11. (T/F) Triads are scored on a curve — ranked against one another so the scores spread out.

12. (MC) During Q&A, when asked a specific question you cannot answer, the disciplined response is:

- A) Give a confident best guess so you don't look unprepared
 - B) "Honestly, I don't know — here's what we'd do to find out"
 - C) Immediately defer the question to the instructor
 - D) Restate your customer + problem to fill the time
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Answer Key

1. **B** — Each prompt validates before the next runs. The validation chain is what keeps the AI Spec honest; running all five in parallel breaks it.
2. **True** — "Senior TPMs use this to draft, not to think." Judgment shapes every section; AI accelerates the typing. AI Spec without validation is fiction.
3. **B** — Compression is selective, not lower-quality: keep what's load-bearing, cut what's cosmetic. Every word earns its place.
4. **B** — 3 happy / 2 sad / 1 weird. Compressed means *fewer* ACs, not weaker ones — still Given/When/Then, same 5-failure-mode discipline.
5. **B** — Trade-off costs, revisit triggers, named invariants, and customer quotes are preserved verbatim; everything else can be synthesized, but hallucinated claims get cut.
6. **False** — Watch for rubber-stamping. The reviewer should find at least one weak claim; "looks fine" without checking against the source is the failure mode.
7. **B** — Each artifact may be fine on its own; the 7 checks ask whether they fit together (e.g., SLO budget fits the sequence diagram, data model supports the API).
8. **False** — Most triads fail 2–3 checks on the first pass. That's expected — that's why the finalization day exists.
9. **B** — Specific (cite the Section), verified, and adds value. "See the TCD" without a section number is a sign the writer never actually verified it.
10. **B** — AI Spec quality at 20% is the highest-weighted dimension — the week's signature artifact carries the most weight.
11. **False** — Score against the rubric, not against other triads. Whole-number scores, no curve: if two triads earn 3.5, both are 3.5.
12. **B** — "Honestly, I don't know — here's what we'd do to find out." Curiosity, not defensiveness; a well-handled "I don't know" beats a hand-wave.